Database Design Project: Auto Insurance Company

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1 Introduction

Insurance companies for vehicles are one of the fundamental pillars of the insurance industry. They strive to provide financial protection and comprehensive coverage for vehicles and their owners in the event of accidents or damages. To enhance their performance and improve their operations, creating a comprehensive and effective database is one of the most important steps.

2 Abstract

The project of creating a database for a vehicle insurance company aims to organize, store, and manage information related to insured vehicles. This includes information related to customers, policies, claims, accidents, maintenance, and other beneficiaries.

The project involves designing and developing a database capable of storing details of insured vehicles, such as technical information, make, model, chassis number, color, and insured value. It also requires recording customer information related to personal data and contact details.

In addition, the database must include a special section for recording policy data and details of the insurance coverage provided for each vehicle and its owner. The policy start and end dates, payment details, and coverage changes can be tracked.

Of course, the project must also include a part for recording claims submitted by customers related to vehicle accidents. Accident details, damages, attached documents, and processing and compensation procedures must be documented.

Developing a database for a vehicle insurance company requires attention to security and professional standards and defining access permissions to ensure the protection of sensitive information.

3 Data Requirements

Based on the initial analysis (Phase 0), the following data requirements have been identified:

- The company has several branches. Each branch is identified by a **Branch Number** and has a **Branch Name**.
- A branch is located at exactly one **Location**. A location can have only one branch.
- A location has a **Serial Number**, **Street Name**, and **City**.
- A branch serves multiple **Customers**.
- Each customer is identified by a unique **Customer Number**. We also store their **Name**, **Date of Birth**, **Age**, **Gender**, and **Phone Number**.
- A customer can have multiple addresses, and an address can be home to multiple customers.
- An address is defined by a **Address ID**, **City**, **Street**, **Building Name**, and **Building Number**.

- The company offers various types of **General Insurance**. Each type has an **Insurance Number**, **Type**, **Start Date**, and **End Date**.
- A customer chooses one type of general insurance for their policy.
- The database stores information about **Drivers** involved in accidents. For each driver, we store their **Name**, **ID Number**, **Date of Birth**, **Driver's License Number**, and **VIN**.
- An **Accident** can affect multiple drivers, and a driver can be affected by multiple accidents.
- An **Accident** record has a **Serial Number**, **Date**, **Location**, and **Description**.
- The **Vehicle** table contains details about the cars insured: **Vehicle Number**, **Name**, **Type**, **Model**, **Year**, **License Number**, and **VIN**.
- A customer owns one or more vehicles, and each vehicle belongs to one customer.
- When an accident occurs, customers file **Claims**. A claim has a **Claim Number**, **Date**, **Amount**, **Description**, and **Status**.
- One accident can lead to multiple claims, but each claim is related to a single accident.
- The policy's **Installments** are tracked. Each installment record has a **Installment Number**, **Due Date**, **Payment Date**, **Amount**, and **Status**.
- Each insured vehicle pays installments, and each installment payment is for one specific insured vehicle.

4 Entity Relationship Diagram (ERD)

The following diagram represents the complete and correct conceptual design of the database, illustrating all entities, their attributes, and the relationships between them. All tables are properly connected according to the business rules.

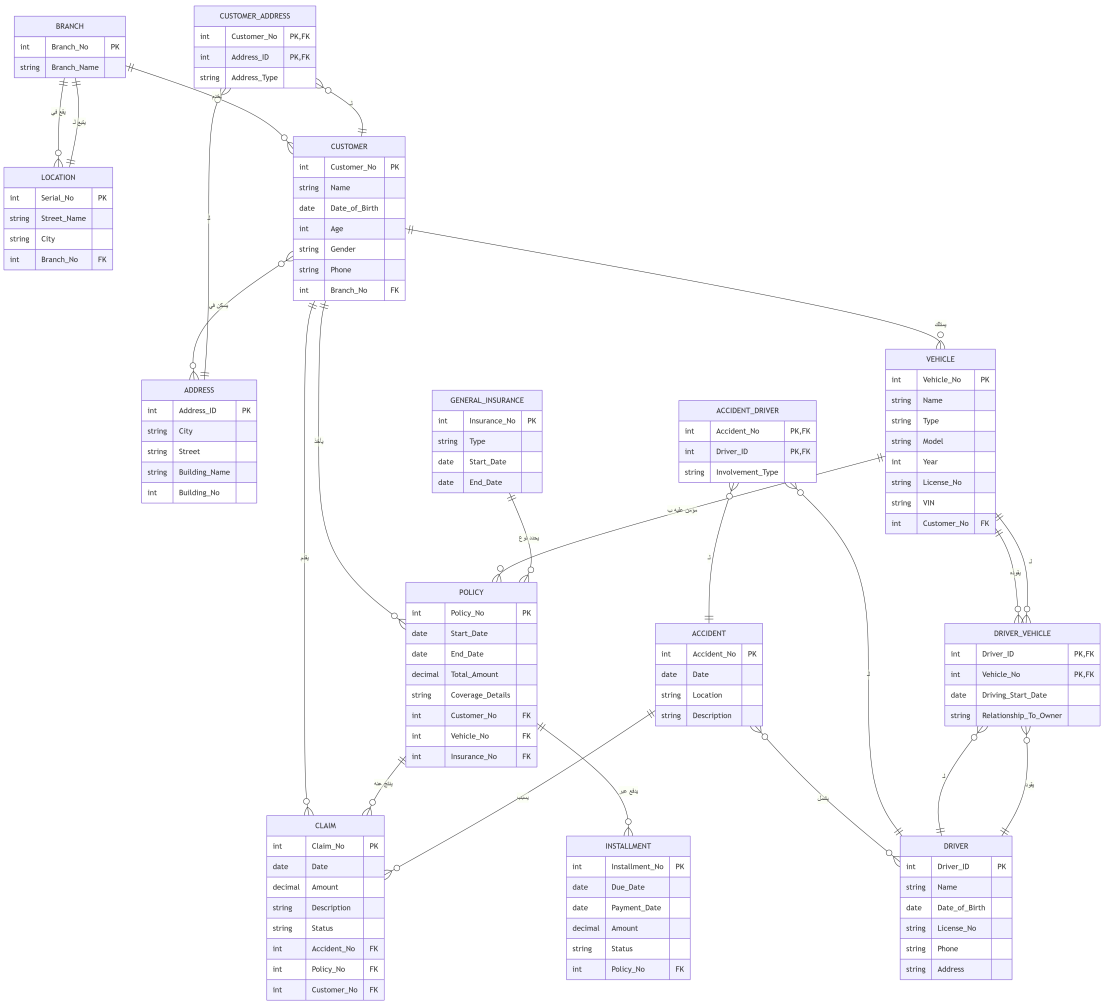

Figure 1: Complete ERD for Auto Insurance Company

5 Tables and Relationships Description

5.1 Main Entities

Table 1: Branch Table

Column	Data Type	Description
Branch_No (PK)	INT	Unique branch identifier
Branch_Name	VARCHAR(100)	Name of the branch

Table 2: Location Table

Column	**Data Type**	**Description**
Serial_No (PK)	INT	Unique location identifier
Street_Name	VARCHAR(100)	Street name
City	VARCHAR(50)	City name
Branch_No (FK)	INT	References BRANCH(Branch_No)

Table 3: Customer Table

Column	**Data Type**	**Description**
Customer_No (PK)	INT	Unique customer identifier
Name	VARCHAR(100)	Customer full name
Date_of_Birth	DATE	Birth date
Age	INT	Customer age
Gender	CHAR(1)	M/F
Phone	VARCHAR(20)	Contact number
Branch_No (FK)	INT	References BRANCH(Branch_No)

Table 4: Address Table

Column	**Data Type**	**Description**
Address_ID (PK)	INT	Unique address identifier
City	VARCHAR(50)	City name
Street	VARCHAR(100)	Street name
Building_Name	VARCHAR(100)	Building name
Building_No	INT	Building number

Table 5: Vehicle Table

Column	**Data Type**	**Description**
Vehicle_No (PK)	INT	Unique vehicle identifier
Name	VARCHAR(100)	Vehicle name
Type	VARCHAR(50)	Vehicle type
Model	VARCHAR(50)	Vehicle model
Year	INT	Manufacturing year
License_No	VARCHAR(20)	License plate number
VIN	VARCHAR(50)	Vehicle Identification Number
Customer_No (FK)	INT	References CUSTOMER(Customer_No)

Table 6: Driver Table

Column	**Data Type**	**Description**
Driver_ID (PK)	INT	Unique driver identifier
Name	VARCHAR(100)	Driver full name
Date_of_Birth	DATE	Birth date
License_No	VARCHAR(20)	Driver's license number
Phone	VARCHAR(20)	Contact number
Address	VARCHAR(200)	Driver address

Table 7: General Insurance Table

Column	**Data Type**	**Description**
Insurance_No (PK)	INT	Unique insurance identifier
Type	VARCHAR(50)	Insurance type
Start_Date	DATE	Coverage start date
End_Date	DATE	Coverage end date

Table 8: Policy Table

Column	**Data Type**	**Description**
Policy_No (PK)	INT	Unique policy identifier
Start_Date	DATE	Policy start date
End_Date	DATE	Policy end date
Total_Amount	DECIMAL(10,2)	Total policy amount
Coverage_Details	TEXT	Coverage description
Customer_No (FK)	INT	References CUSTOMER(Customer_No)
Vehicle_No (FK)	INT	References VEHICLE(Vehicle_No)
Insurance_No (FK)	INT	References GENERAL_INSURANCE(Insurance_No)

Table 9: Accident Table

Column	**Data Type**	**Description**
Accident_No (PK)	INT	Unique accident identifier
Date	DATE	Accident date
Location	VARCHAR(200)	Accident location
Description	TEXT	Accident description

Table 10: Claim Table

Column	Data Type	Description
Claim_No (PK)	INT	Unique claim identifier
Date	DATE	Claim filing date
Amount	DECIMAL(10,2)	Claim amount
Description	TEXT	Claim description
Status	VARCHAR(20)	Claim status
Accident_No (FK)	INT	References ACCIDENT(Accident_No)
Policy_No (FK)	INT	References POLICY(Policy_No)
Customer_No (FK)	INT	References CUSTOMER(Customer_No)

Table 11: Installment Table

Column	Data Type	Description
Installment_No (PK)	INT	Unique installment identifier
Due_Date	DATE	Payment due date
Payment_Date	DATE	Actual payment date
Amount	DECIMAL(10,2)	Installment amount
Status	VARCHAR(20)	Payment status
Policy_No (FK)	INT	References POLICY(Policy_No)

5.2 Relationship Tables (Junction Tables)

Table 12: Customer-Address Junction Table

Column	Data Type	Description
Customer_No (PK, FK)	INT	References CUSTOMER(Customer_No)
Address_ID (PK, FK)	INT	References ADDRESS(Address_ID)
Address_Type	VARCHAR(50)	Primary/Secondary/Business

Table 13: Accident-Driver Junction Table

Column	Data Type	Description
Accident_No (PK, FK)	INT	References ACCIDENT(Accident_No)
Driver_ID (PK, FK)	INT	References DRIVER(Driver_ID)
Involvement_Type	VARCHAR(50)	Responsible/Affected/Witness

Table 14: Driver-Vehicle Junction Table

Column	Data Type	Description
Driver_ID (PK, FK)	INT	References DRIVER(Driver_ID)
Vehicle_No (PK, FK)	INT	References VEHICLE(Vehicle_No)
Driving_Start_Date	DATE	When driver started driving this vehicle
Relationship_To_Owner	VARCHAR(50)	Owner/Family Member/Authorized Driver

6 Relationship Summary

- **BRANCH - LOCATION**: One-to-One relationship. Each branch has exactly one location.
- **BRANCH - CUSTOMER**: One-to-Many relationship. One branch serves many customers.
- **CUSTOMER - ADDRESS**: Many-to-Many relationship resolved by CUSTOMER_ADDRESS table.
- **CUSTOMER - VEHICLE**: One-to-Many relationship. One customer owns many vehicles.
- **CUSTOMER - POLICY**: One-to-Many relationship. One customer can have multiple policies over time.
- **VEHICLE - POLICY**: One-to-Many relationship. One vehicle can have multiple policies (renewals).
- **VEHICLE - DRIVER**: Many-to-Many relationship resolved by DRIVER_VEHICLE table.
- **ACCIDENT - DRIVER**: Many-to-Many relationship resolved by ACCIDENT_DRIVER table.
- **ACCIDENT - CLAIM**: One-to-Many relationship. One accident can generate multiple claims.
- **POLICY - CLAIM**: One-to-Many relationship. One policy can generate multiple claims.
- **POLICY - INSTALLMENT**: One-to-Many relationship. One policy has multiple installments.

7 Conclusion

This database design provides a comprehensive solution for managing all aspects of an auto insurance company. It ensures data integrity, eliminates redundancy, and maintains proper relationships between all entities. The design follows database normalization principles and includes all necessary junction tables to handle many-to-many relationships correctly.

8 Project Resources

The project repository and documentation can be found at the following link:

GitHub Repository

`https://github.com/layan_Buirat`

— End of Report —